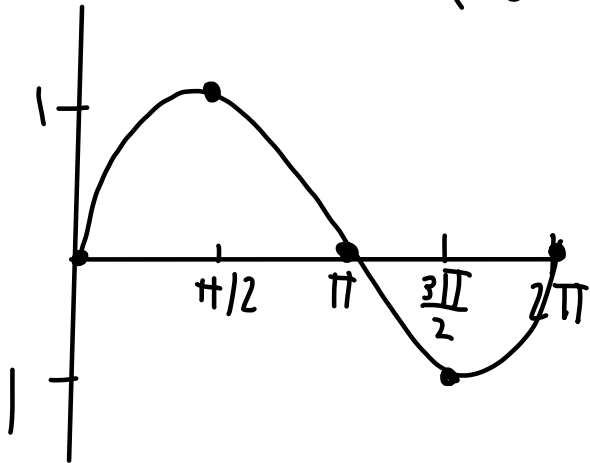


Neural Networks are great at learning nonlinear complex relationships.

$f(x) = \sin x$ (true distribution, code has some random noise to simulate irreducible error.)



We sample 100 evenly spaced points. We feed the x into the NW and y -noisy is the target.

We have 1 feature with 100 examples so dimensions of x and y -noisy have to be $(1, 100)$ in order to follow the (features, examples) convention. $[0.1, 0.5, \dots, 0.4]$

Our model will have an input layer, 2 hidden layers, and an output layer. We note that the input is a single neuron, the hidden layers are 8 neurons, and the output is a single neuron. The formula for # of weight matrices is the following: number of layers - 1. Also, for matrix shape, the pair is (neurons out, neurons in).

This is because we picked (features, examples) convention so matrix multiplication will be $w \cdot x + b$. As for the bias b term, the rule is (neurons-out, 1) due to broadcasting (adding to each column).

Input (X): (1, 100)

$$\begin{matrix} W_1: (8, 1) \\ B_1: (8, 1) \end{matrix} \} (8, 1) \cdot (1, 100) \Rightarrow (8, 100) \Rightarrow \text{shape of } z_1$$

$$\begin{matrix} W_2: (8, 8) \\ B_2: (8, 1) \end{matrix} \} (8, 8) \cdot (8, 100) \Rightarrow (8, 100) \Rightarrow \text{shape of } z_2$$

$$\begin{matrix} W_3: (1, 8) \\ B_3: (1, 1) \end{matrix} \} (1, 1) \cdot (8, 100) \Rightarrow (1, 100) \Rightarrow \text{shape of } z_3$$

Each layer's output size determines next layers input.

$$X \rightarrow z_1 \xrightarrow{w_1 x_1 + b_1} a_1 \xrightarrow{\text{relu}(a_1)} z_2 \xrightarrow{w_2 a_1 + b_2} a_2 \xrightarrow{\text{relu}(z_2)} z_3 \xrightarrow{w_3 a_2 + b_3} L$$

The $\sqrt{2}$ comes from kaiming initialization in the code.
#input connections.

Forward-prop(X) (1, 100)

$$z_1 = W_1 X_1 + b_1 \quad (8, 100)$$

$$a_1 = \text{relu}(z_1) \quad (8, 100)$$

$$z_2 = W_2 a_1 + b_2 \quad (8, 100)$$

$$a_2 = \text{relu}(z_2) \quad (8, 100)$$

$$z_3 = W_3 a_2 + b_3$$

return z_1, a_1, z_2, a_2, z_3

no activation after, just identity since we are doing regression. Classification would have sigmoid function.

Loss function: $\frac{1}{2n} \sum_{i=1}^n (z_3[i] - y[i])^2$

Annotations:
 - $\frac{1}{2n}$ makes deriv easier
 - $z_3[i]$ predicted
 - $y[i]$ actual

$$L = \frac{1}{2n} \sum_{i=1}^n z_3(i)^2 - 2z_3(i)y(i) + y(i)^2$$

$$\frac{\partial L}{\partial z_3}[i] = \frac{1}{n} \left(\sum_{i=1}^n z_3(i) - y(i) \right) \Rightarrow (1, 100)$$

$$\frac{\partial L}{\partial w_3} = \frac{\partial L}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_3} \quad \left\{ \frac{\partial L}{\partial w_3} \text{ must match } w_3 \text{ dimensions of } (1, 8) \right.$$

$$z_3 = w_3 a_2 + b_3$$

$$\frac{\partial z_3}{\partial w_3}[i] = a_2[i] \Rightarrow (8, 100)$$

We have $\frac{\partial L}{\partial z_3}$ with shape $(1, 100)$ and $\frac{\partial z_3}{\partial w_3}$ of shape $(8, 100)$. We want to match w_3 dimensions of $(1, 8)$ for $\frac{\partial L}{\partial w_3}$.

$$\frac{\partial L}{\partial w_3} = \frac{\partial L}{\partial z_3} \cdot a_2^T$$

Dimensions: $(1, 100) \cdot (100, 8)$

We will now compute loss with respect to bias term (b_3 in this case).

$$\frac{\partial L}{\partial b_3} = \frac{\partial L}{\partial z_3} \cdot \frac{\partial z_3}{\partial b_3}$$

$$\frac{\partial L}{\partial z_3} [i] = \frac{1}{n} \left(\sum_{i=1}^n z_3(i) - y(i) \right) \Rightarrow (1, 100)$$

$$z_3 = w_3 a_2 + b_3 \quad \text{We want } \frac{\partial L}{\partial b_3} \text{ with shape } (1, 1).$$

$$\frac{\partial z_3}{\partial b_3} = 1 \Rightarrow (1, 1) \quad \text{We can just collapse } \frac{\partial L}{\partial z_3}.$$

We now move onto the next layer.

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial w_2}$$

$$\frac{\partial L}{\partial w_2} = \underbrace{\frac{\partial L}{\partial z_3}}_{\substack{\text{already have, shape } (1, 100)}} \cdot \frac{\partial z_3}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial w_2}$$

$$z_3 = w_3 a_2 + b_3$$

$$\frac{\partial z_3}{\partial a_2} [i] = w_3 [i] \Rightarrow \text{shape } (1, 8)$$

$$\text{relu}(z_2) = \begin{cases} z_2, & z_2 > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\frac{\partial a_2}{\partial z_2} = \text{relu}'(z_2) = \begin{cases} 1, & z_2 > 0 \\ 0, & \text{otherwise} \end{cases} \Rightarrow \text{shape } (8, 100)$$

$$z_2 = w_2 a_1 + b_2$$

$$\frac{\partial z_2}{\partial w_2} [i] = a_1 [i] \Rightarrow \text{shape } (8, 100)$$

We need $\frac{\partial L}{\partial w_2}$ to match w_2 shape (8, 8).

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial z_3} \cdot w_3 \cdot \text{relu}'(z_2) \cdot a_2$$

$\downarrow$ $\downarrow$ $\downarrow$
 (1, 100) (1, 8) (8, 100) (8, 100)

$$= \frac{\partial L}{\partial a_2} \cdot \text{relu}'(z_2) \cdot a_2$$

$\frac{\partial L}{\partial a_2}$ should have same shape as a_2 which has shape (8, 100).

Note that $\frac{\partial L}{\partial a_2} = \frac{\partial L}{\partial z_3} \cdot w_3$. w_3 has the shape (1, 8)

so we do $w_3^T \cdot \frac{\partial L}{\partial z_3}$.

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial a_2} \cdot \text{relu}'(z_2) \cdot a_2$$

$\downarrow$ $\downarrow$ $\downarrow$
 (8, 100) (8, 100) (8, 100)

needs to be (8, 8)

Since Relu transforms per element and $\text{relu}(z_2)$ only relies on z_2 , we do the hadamard of $\frac{\partial L}{\partial a_2}$ and

$\text{relu}'(z_2)$ in order to get: $\frac{\partial L}{\partial z_2} = \frac{\partial L}{\partial a_2} \odot \text{relu}'(z_2)$ in order

to get shape (8, 100). Finally, we do $\frac{\partial L}{\partial z_2} \cdot a_2^T$ to get matching shape (8, 8) of w_2 .

Now we want to calculate $\frac{\partial L}{\partial b_2}$.

$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial z_2} \cdot \frac{\partial z_2}{\partial b_2}$$

$\Rightarrow \text{shape}(8, 100)$

$$z_2 = w_2 a_1 + b_2$$

$$\frac{\partial z_2}{\partial b_2} = 1$$

$\Rightarrow$ already computed. We want to get shape $(8, 1)$ which means we need collapse the number of columns of $\frac{\partial L}{\partial z_2}$ into 1 and maintain number of neurons so we add alongside axis=2.